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*          STAAD.Pro
*          Version 2007   Build 04
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date=   APR 1, 2013
*          Time=   15:35:23
*
*          USER ID: Parsons Binckerhoff
*****

1. STAAD SPACE
INPUT FILE: 130320 Tie Beam, TB1.STD
2. START JOB INFORMATION
3. ENGINEER DATE 22-NOV-11
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 3 0 0 0; 4 7.6 0 0; 5 -6.2 0 0; 6 13.2 0 0
9. MEMBER INCIDENCES
10. 3 3 4; 4 3 5; 5 4 6
11. DEFINE MATERIAL START
12. ISOTROPIC CONCRETE
13. E 2.17185E+007
14. POISSON 0.17
15. DENSITY 23.5616
16. ALPHA 1E-005
17. DAMP 0.05
18. END DEFINE MATERIAL
19. MEMBER PROPERTY AMERICAN
20. 3 TO 5 PRIS YD 0.8 ZD 0.7
21. CONSTANTS
22. MATERIAL CONCRETE ALL
23. SUPPORTS
24. 3 TO 5 PINNED
25. 6 FIXED BUT MY MZ KMX 0.001
26. LOAD 1 LOADTYPE NONE TITLE DEAD LOAD
27. SELFWEIGHT Y -1.4
28. MEMBER LOAD
29. 3 TO 5 CON GY -1062.1
30. 3 TO 5 CON GZ 19
31. 3 TO 5 CON GX 200
32. 3 TO 5 CMOM GY 0.05
33. 3 CMOM GX 58
34. 3 TO 5 CMOM GZ 500
35. 3 TO 5 UNI GZ 21
36. 3 5 CON GY -140
37. 3 5 CON GZ 10
38. 3 5 CON GX 15
39. 3 5 CMOM GX 80
40. 3 5 CMOM GZ 15
41. *****
42. LOAD COMB 2 1.0DL
43. 1 1.0
44. PERFORM ANALYSIS

P R O B L E M   S T A T I S T I C S
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NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =      4/      3/      4

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER
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ORIGINAL/FINAL BAND-WIDTH= 2/ 2/ 9 DOF
TOTAL PRIMARY LOAD CASES = 1, TOTAL DEGREES OF FREEDOM = 12
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 20378.6 MB

45. LOAD LIST 2
46. PRINT JOINT DISPLACEMENTS ALL

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
3	2	0.000E+00	0.000E+00	0.000E+00	3.805E+03	-2.338E-05	-7.947E-05
4	2	0.000E+00	0.000E+00	0.000E+00	3.805E+03	3.844E-05	-3.962E-04
5	2	0.000E+00	0.000E+00	0.000E+00	3.805E+03	-2.512E-04	-2.450E-03
6	2	0.000E+00	0.000E+00	0.000E+00	3.805E+03	2.001E-04	1.901E-03

***** END OF LATEST ANALYSIS RESULT *****

47. PRINT MEMBER FORCES ALL

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
3	2	3	-107.50	741.78	-95.07	0.00	127.49	1268.49
		4	-107.50	600.71	-93.53	-138.00	-121.69	-1247.40
4	2	3	100.00	712.26	95.15	0.00	-127.49	1268.49
		5	100.00	464.36	54.05	0.00	0.00	0.00
5	2	4	-107.50	967.49	-95.04	138.00	121.69	1247.40
		6	-107.50	338.06	-51.56	-218.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

48. PRINT SUPPORT REACTION ALL

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
3	2	-207.50	1454.05	-190.22	0.00	0.00	0.00
4	2	-215.00	1568.19	-188.57	0.00	0.00	0.00
5	2	-100.00	464.36	-54.05	0.00	0.00	0.00
6	2	-107.50	338.06	-51.56	-218.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

49. LOAD LIST ALL
 50. *START CONCRETE DESIGN
 51. *CODE CP65
 52. *DESIGN BEAM 3 TO 5
 53. *END CONCRETE DESIGN
 54. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= APR 1,2013 TIME= 15:35:23 ****

 * For questions on STAAD.Pro, please contact *
 * Research Engineers Offices at the following locations *
 * *
 * Telephone Email *
 * USA: +1 (714)974-2500 support@bentley.com *
 * CANADA +1 (905)632-4771 detech@odandetech.com *
 * UK +44(1454)207-000 support@bentley.com *
 * NORWAY +47 67 57 21 30 staa@edr.no *
 * SINGAPORE +65 6225-6158 support@bentley.com *
 * INDIA +91(033)4006-2021 support@bentley.com *
 * JAPAN +81(03)5952-6500 eng-eye@crc.co.jp *
 * CHINA +86(411)8479-1166 support@bentley.com *
 * THAILAND +66(0)2645-1018/19 support@bentley.com *
 * *
 * North America support@bentley.com *
 * Europe support@bentley.com *
 * Asia support@bentley.com *
